

Phylogenetics - III

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How to construct molecular phylogenetics tree?

Molecular phylogenetics tree construction can be divided into five steps;

1. Choosing molecular markers
2. Performing Multiple Sequence Alignment (MSA)
3. Choosing a model of evolution
4. Determining a tree building method
5. Assessing tree reliability



1. Choice

Nucleot

- For studying cl since they evol
- Use of mtDNA, i
- Sequence/codo

Mitochondrial code

		Second letter				
		U	C	A	G	
First letter	U	Phe	Ser	Tyr	Cys	U
		Phe	Ser	Tyr	Cys	C
		Leu	Ser	Stop	(Stop) Trp	A
		Leu	Ser	Stop	Trp	G
	C	Leu	Pro	His	Arg	U
		Leu	Pro	His	Arg	C
		Leu	Pro	Gln	Arg	A
		Leu	Pro	Gln	Arg	G
	A	Ile (Met)	Thr	Asn	Ser	U
		Ile	Thr	Asn	Ser	C
		(Ile) Met	Thr	Lys	(Arg) Stop	A
		Ile	Thr	Lys	(Arg) Stop	G
	G	Val	Ala	Asp	Gly	U
		Val	Ala	Asp	Gly	C
		Val	Ala	Glu	Gly	A
		Val	Ala	Glu	Gly	G

Universal code

		Second letter							
		U		C		A		G	
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC		UCC		UAC		UGC		C
	UUA	UCA	UAA		UGA	Stop	A		
	UUG	UCG	UAG		UGG	Trp	G		
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	CGA	A		
	CUG		CCG		CAG	CGG	G		
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC		ACC		AAC		AGC		C
	AUA		ACA		AAA	AGA	A		
	AUG		ACG		AAG	AGG	G		
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	GGA	A		
	GUG		GCG		GAG	GGG	G		

- Different genetic code of mtDNA



- Alignment sensitivity! High signal to noise ratio in protein alignments
- Biological irrelevance when gaps are introduced in protein sequence alignment.
- *Synonymous* and *nonsynonymous* substitutions in DNA sequences



2. Alignment

- Most critical step in constructing a phylogenetic tree.
 - Correct alignment → genealogically relevant tree
 - Wrong alignment → wrong tree
- Multiple alignment methods should be used, inspected and compared carefully.



3. Choice of evolution model

	A	G	C	T
A		α	α	α
G			α	α
C				α
T				

Jukes-Cantor model

	A	G	C	T
A		α	β	β
G			β	β
C				α
T				

Kimura model



4. Tree building method

Three main methods;

1. Distance-based
2. Maximum parsimony
3. Maximum likelihood



4.1. Distance based method

- Calculate the **distance** between each pair of sequence
- Sequence pairs that have the smallest changes between them are termed **neighbors**
- **Join** the neighbors by a **node** and repeat the process



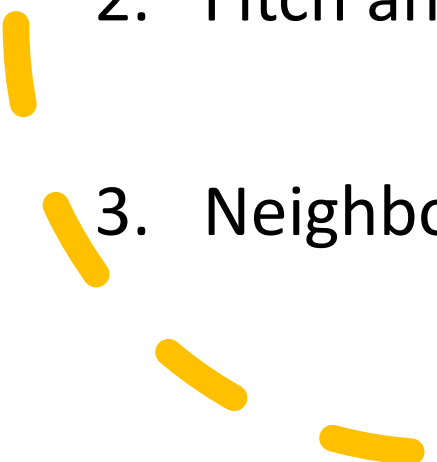


Many variations but we will talk about three widely used ones,

1. Unweighted Pair Group Method with Arithmetic Mean (UPGMA)

2. Fitch and Margoliash

3. Neighbor joining



4.1.1. UPGMA

- Simple method
- Assumes a 'molecular clock'
 - Rate of change is constant in sister species
 - Hence branch lengths are equal
- Starts from a distance matrix
- Pairs up the most closely related taxa
- Treat this pair as new taxa



A: ATCGTGGTACTG

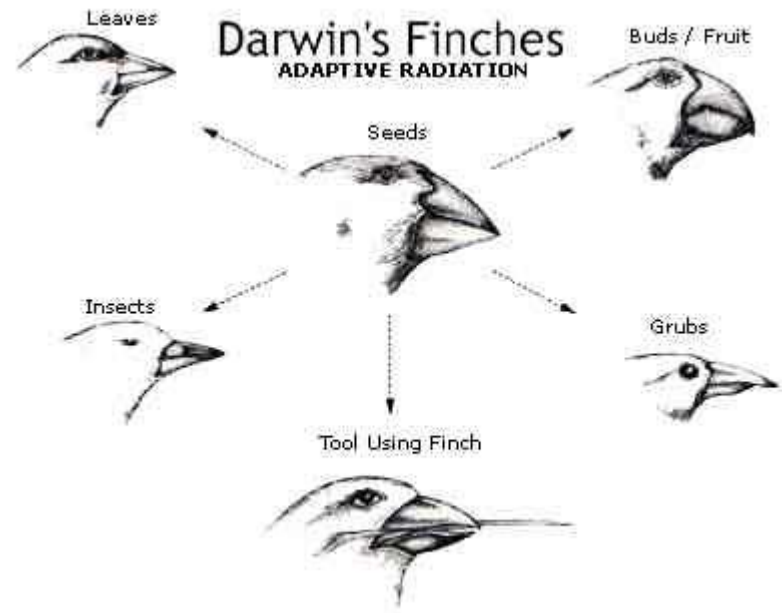
B: CCGGAGAACTAG

C: AACGTGCTACTG

D: ATGGTGAAAGTG

E: CCGGAAAACCTTG

F: TGGCCCTGTATC



[[A
C
B
E]]

	A	B	C	D	E	F
A	-	9	2	4	9	11
B	-	-	9	6	2	11
C	-	-	-	5	9	11
D	-	-	-	-	6	10
E	-	-	-	-	-	10
F	-	-	-	-	-	-



	A	B	C	D	E	F
A	-	9	2	4	9	11
B	-	-	9	6	2	11
C	-	-	-	5	9	11
D	-	-	-	-	6	10
E	-	-	-	-	-	10
F	-	-	-	-	-	-



	A/C	B	<u>D</u>	E	F
A/C	-	9	4.5	9	11
B	-	-	6	2	11
D	-	-	-	6	10
E	-	-	-	-	10
F	-	-	-	-	-

$$\begin{bmatrix} A \\ C \end{bmatrix}$$

$$\frac{9+9}{2} = 9$$

$$\begin{bmatrix} B \\ E \end{bmatrix}$$

$$\frac{4+5}{2} = 4.5$$



	A/C	B	D	E	F
A/C	-	<u>9</u>	4.5	<u>9</u>	11
B	-	-	<u>6</u>	2	11
D	-	-	-	6	10
E	-	-	-	-	10
F	-	-	-	-	-



$$\frac{9+9}{2} = 9$$

$$\frac{6+6}{2} = 6$$

$$\frac{11+10}{2} = 10.5$$

✓	A/C	B/E	D	F
A/C	-	9	4.5	11
B/E	-	-	6	10.5
D	-	-	-	10
F	-	-	-	-

[A
 [C
 [D
 [B
 [E



	A/C	B/E	D	F
A/C	-	9	4.5	11
B/E	-	-	6	10.5
D	-	-	-	10
F	-	-	-	-



	A/C/D	B/E	F
A/C/D	-	7.5	10.5
B/E	-	-	10.5
F	-	-	-

